

B Future Work/Limitations

Several limitations constrain the generalizability of our findings. First, our dataset consists of synthetic, templated prompts rather than naturalistic conversational contexts, potentially limiting ecological validity. The reliance on log-probabilities rather than actual text generation may not fully capture how rebound manifests in practical deployment scenarios where models produce complete responses. Additionally, our analysis focused exclusively on English and single-token target concepts, leaving open questions about cross-linguistic generalization and multi-word forbidden phrases. Our experimental design, while systematic, employed only three distractor types (semantic, syntactic, repetition) and may not capture the full spectrum of cognitive load conditions that occur in real-world usage. The artificial nature of our "do not mention X" instructions also differs from the more nuanced content policies and safety filters deployed in production systems, which often involve complex multi-step reasoning about appropriateness rather than simple lexical prohibition.

Future research should address several critical directions. First, mechanistic interpretability work is needed to identify the specific circuits responsible for ironic rebound, potentially building on existing analyses of copy-suppression heads to understand how negation instructions are processed and how they interact with attention mechanisms across different context lengths. Second, studies should explore more naturalistic settings, including conversational contexts where negation emerges organically rather than through explicit instructions.

The anomalous behavior of GPT-OSS-20B, which contradicts the observed scaling trends, warrants further investigation. We hypothesize this may stem from its specific fine-tuning process, which was based on a larger proportion of synthetic data than the other models we observed, or from architectural variations not present in other models of similar scale.

Investigation of mitigation strategies represents another crucial avenue. If negation-based filters are inherently fragile, alternative approaches such as positive framing ("focus on Y instead of X") or multi-step reasoning chains may prove more robust. Additionally, the scaling trends we observed suggest that architectural modifications or training procedures might be developed to reduce rebound susceptibility while preserving the enhanced polarity discrimination capabilities of larger models. Finally, direct behavioral comparisons between human and model rebound patterns could clarify whether observed effects reflect genuine cognitive parallels or merely superficial statistical regularities, informing both our understanding of language model cognition and the development of more human-aligned AI systems.

B.1 Regression Results

Coefficient	Coef.	Std.Err.	P> z	[0.025, 0.975]
GPT-NeoX-20B	0.366	0.051	0.000	[0.266, 0.466]
GPT-OSS-20B	-10.692	0.051	0.000	[-10.792, -10.592]
GPT-2-Small	-3.446	0.051	0.000	[-3.546, -3.346]
LFM2-350M	-3.995	0.051	0.000	[-4.095, -3.895]
Llama-3-8B-Instruct	-4.822	0.051	0.000	[-4.922, -4.722]
OPT-2.7B	-2.398	0.051	0.000	[-2.498, -2.298]
Pythia-410M	-4.789	0.051	0.000	[-4.888, -4.689]
Qwen3-14B	-1.643	0.051	0.000	[-1.743, -1.544]

Table 1: Mixed-effects regression results for the surprisal difference $\Delta(\text{bits})$.

Notes: *R-squared*: 0.564, *Adjusted R-squared*: 0.564, *F-statistic*: 2563, *Prob (F-statistic)*: 4.79e-86. Standard errors are robust to cluster correlation.

Coefficient	Coef.	Std.Err.	P> z	[0.025, 0.975]
GPT-NeoX-20B	0.3661	0.469	0.435	[-0.553, 1.286]
GPT-OSS-20B	-10.6916	0.521	0.000	[-11.712, -9.671]
GPT-2-Small	-3.4459	0.518	0.000	[-4.460, -2.431]
LFM2-350M	-3.9949	0.449	0.000	[-4.876, -3.114]
Llama-3-8B-Instruct	-4.8224	0.569	0.000	[-5.938, -3.707]
OPT-2.7B	-2.3977	0.473	0.000	[-3.324, -1.471]
Pythia-410M	-4.7886	0.456	0.000	[-5.682, -3.895]
Qwen3-14B	-1.6435	0.458	0.000	[-2.541, -0.746]

Table 2: OLS Regression Results for the surprisal difference $\Delta(\text{bits})$.

B.2 Detailed Results: Irony Rebound vs. Load

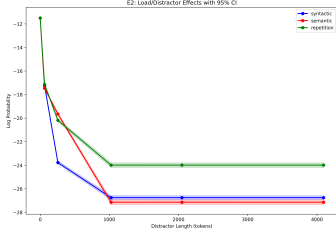


Figure B2.1: GPT-2-Small

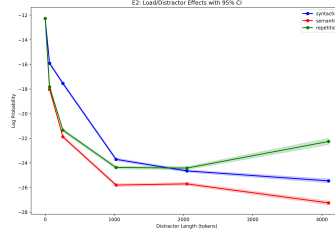


Figure B2.2: LFM2-350M

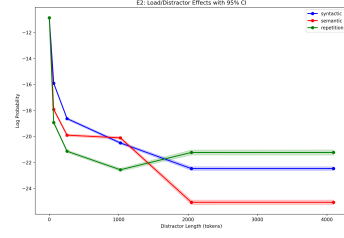


Figure B2.3: Pythia-410M

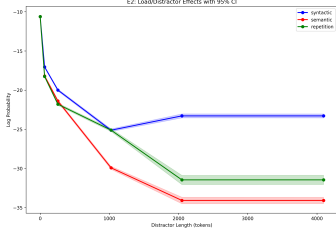


Figure B2.4: Bloom-560M

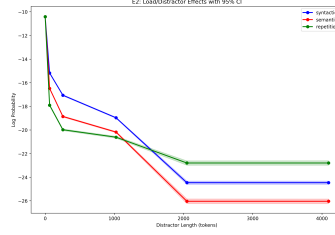


Figure B2.5: OPT-2.7B

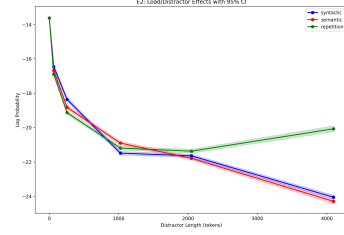


Figure B2.6: Llama-3-8B-Instruct

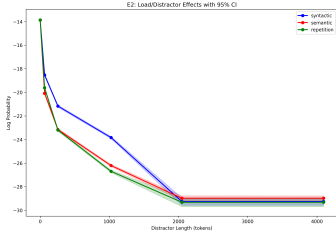


Figure B2.7: Qwen3-14B

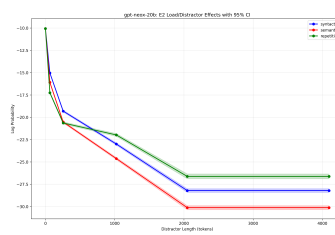


Figure B2.8: GPT-NeoX-20B

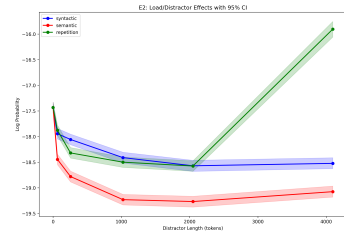


Figure B2.9: GPT-OSS-20B

Model	Distractor Type	Peak Δ (bits)	L_{50}	AUC Δ
GPT-2 Small	Semantic	18.2	0.09	1.73
	Syntactic	17.8	0.05	0.91
	Repetition	13.9	0.06	0.98
LFM2-350M	Semantic	17.4	0.06	2.09
	Syntactic	16.4	0.14	2.49
	Repetition	10.4	0.04	0.38
Pythia-410M	Semantic	15.4	0.06	2.89
	Syntactic	13.1	0.06	1.62
	Repetition	9.1	0.04	0.21
Bloom-560M	Semantic	28.4	0.13	4.26
	Syntactic	13.7	0.05	0.88
	Repetition	24.6	0.13	4.35
OPT-2.7B	Semantic	18.2	0.20	3.61
	Syntactic	16.8	0.23	3.50
	Repetition	12.5	0.05	1.49
Llama-3-8B-Instruct	Semantic	13.2	0.15	2.83
	Syntactic	13.0	0.14	2.58
	Repetition	6.9	0.05	0.31
Qwen3-14B	Semantic	17.3	0.06	2.33
	Syntactic	18.8	0.18	3.64
	Repetition	18.1	0.06	2.35
GPT-NeoX-20B	Semantic	24.6	0.12	4.06
	Syntactic	22.6	0.12	3.80
	Repetition	18.7	0.06	3.00
GPT-OSS-20B	Semantic	1.6	0.05	0.09
	Syntactic	1.2	0.09	0.15
	Repetition	-2.5	0.95	0.00

Table 3: Summary of rebound dynamics across models and distractor types. Peak Δ indicates the maximum rebound in surprisal bits (higher = stronger rebound). L_{50} is the normalized load at which rebound falls to half of its peak (higher = more persistent).

C Further Attention Head Analysis

C.1 Most Attention Heads Have Little Effect on Rebound

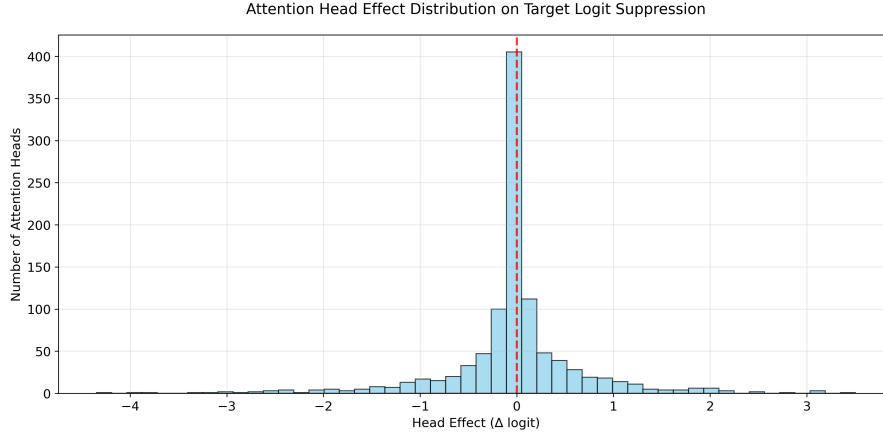


Figure C1.1: Distribution of attention head effects on forbidden token probability. Most heads have minimal impact, but a sparse few create strong suppression or amplification effects.

We tested all attention heads in Llama 3 8B Instruct, a 32-layer model (over 1,000 heads total) that showed a clear rebound effect in our initial measurement. We found that the vast majority of heads had essentially no impact on whether the model produces forbidden tokens. However, a small number of heads had large effects, either strongly suppressing forbidden words (negative effects) or amplifying them (positive effects). Figure C1.1 shows this distribution, with most heads clustered near zero but a few outliers reaching extreme values of up to 4 logits in either direction.

This pattern suggests that ironic rebound emerges from the interaction of a relatively small number of specialized components, rather than being distributed across the entire model.

C.2 Summary of Internal Mechanisms

Our circuit analysis reveals that ironic rebound is not simply a failure of language models to follow instructions. Instead, it emerges from identifiable internal mechanisms: early layers implement suppression as intended, but a sparse set of amplifying heads in middle layers partially reverses these effects. Across the full model architecture, 26.7% of heads contribute to rebound, 29.7% to suppression, and 43.7% remain functionally neutral. This suggests that more robust negation following might be achievable through precision modifications rather than holistic architectural changes.

Moreover, the dynamics we observe raise questions about whether different types of distractors have varying effects on rebound. Semantic distractors may engage the same middle-layer heads responsible for amplification, while repetitive text might strengthen the early suppression circuits. Understanding these internal mechanisms opens new avenues for both explaining and potentially controlling ironic rebound in future language models.